

Todo list

	piano scores	3
	options (also for packages)	3
	indent	3
	aumentar longitud	3
	monotone triplets sliding beam	5
	bar on grupetto	6
	ta taca beams	6
	maybe fix this	7
	fix, add swap	9
	Fix angle	9
	crescendo	9
	articulation	9
	explain spaces	10
	Tempo command	11
	beautiful code	12

The `mousik` package*

Celia Rubio Madrigal†

July 1, 2020

Abstract

This is a music-related package focused on writing music fragments. It works specially well with small examples, such as excerpts for theoretical articles. It is easier than other music writing methods in $\text{\LaTeX}$ because it is based on the `tikz` package.

Keywords

mousik, music, music theory, music notation, music writing, score, sheet, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key

Contents

		3.6.2	<code>\Key</code>	8
		3.6.3	<code>\NoKey</code>	8
1	Introduction	3	3.7 Other symbols	9
			3.7.1 <code>\Dot</code>	9
2	The <code>mousik</code> environment	3	3.7.2 <code>\Tied</code>	9
			3.7.3 The slurred environment	9
3	Available commands	4	3.8 <code>\Text</code> and <code>\Dynamics</code>	10
	3.1 <code>\Bar</code>	4		
	3.2 <code>\Space</code>	4		
	3.3 <code>\Clef</code>	4	4	Observations
	3.4 <code>\TimeSig</code>	5		
	3.5 Notes and rests	5	4.1	Overflow of lines
			4.2	How do spaces work?
	3.5.1 <code>\Note</code>	5		
	3.5.2 <code>\Rest</code>	6	5	A very long and complete example
	3.5.3 <code>\Chord</code>	6		11
	3.6 Keys and accidentals	7	6	The package code
	3.6.1 <code>\Acc</code>	7		12

*This document corresponds to `mousik` v0.1, dated 2020/07/01.

†Email: celirubio.m@gmail.com

1 Introduction

There are already several ways to set music using L^AT_EX. As stated by the `abc` package, “*notably Music_{TE}X and Lilypond*”, as well as `abc` per se. Some of them are too difficult to use for not too complicated music; others do not work at all.

The `mousik` package is perfect for short, one line music fragments. It may be used for longer scores, but there are more adaptable and lighter packages than this one. However, in spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

piano scores

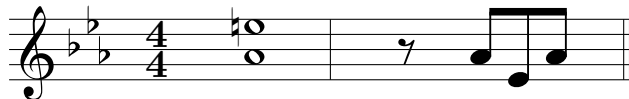
options (also for packages)

indent

aumentar
longitud

2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment.



```
\begin{mousik}
  \Clef{2}
  \Key{-3}
  \TimeSig{4}{4}

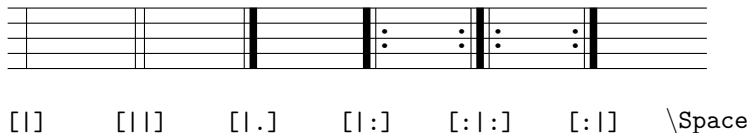
  \Note{1}{5}
  \Chord
  \Acc{0}{9}
  \Note{1}{9}
  \Bar

  \Rest{8}
  \Note{8}{5,2,5}
  \Bar[|. ]
\end{mousik}
```

3 Available commands

3.1 `\Bar`

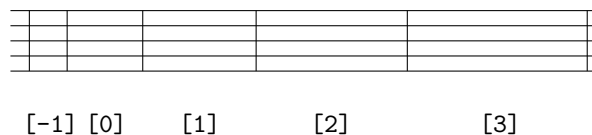
The `\Bar` command prints a bar on the score. It has an optional argument depending on the type of bar, with default `|`, the single bar (`\Bar = \Bar[]`).



The only way to end a line in the score is with a bar or a space. Any other way may cause visual errors (see subsection 4.1).

3.2 `\Space`

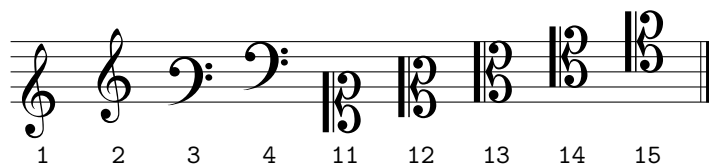
The `\Space` command is not only used when no bar is needed. It may also be used to input *extra* space between symbols. The default space is 2 (which is the default space between symbols in general), but it may be any whole number (see subsection 4.2).



3.3 `\Clef`

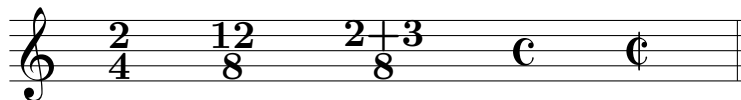
Clefs might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

- `\Clef{1}` and `\Clef{2}` are Treble Clefs in the first and second lines.
- `\Clef{3}` and `\Clef{4}` are Bass Clefs in the third and fourth lines.
- `\Clef{11}`, `\Clef{12}`, `\Clef{13}`, `\Clef{14}` and `\Clef{15}` are Alto Clefs in each of the five lines.



3.4 `\TimeSig`

Time signatures might be used at any moment. `\TimeSig{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets. Also `\TimeSig{C}{}` will produce a *common time* symbol and `\TimeSig{C|}{}` will produce an *alla breve* symbol.

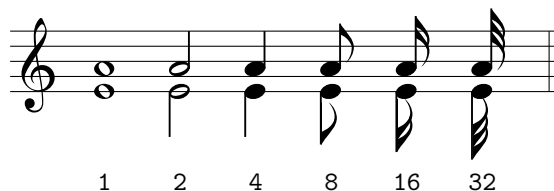


3.5 Notes and rests

3.5.1 `\Note`

The `\Note` command prints a note on the score. It accepts 2 arguments.

The first argument is the duration of the note, from one of these:



The second argument is the absolute height. For example, these are the notes from 0 to 9:



The `\Note` command might also print several notes tied together with a beam. Two notes will be tied together with an oblique beam, but three or more will have a straight beam.



monotone triplets sliding beam

The notes go upside down when their height (or maximum height) is bigger than 6. However, for manual control, the `\Swap` command might be used before

the note, as seen above in the second group.

To write the grupetto number of a tuplet, the `\Grupetto` command might be used *after* the note. It has two arguments, an optional one and a compulsory one. The optional one is the written number, which is automatically calculated by default. The compulsory one is the height: `\Grupetto{-5}`.

ta tacata beams

bar on grupetto

3.5.2 `\Rest`

The `\Rest` command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



3.5.3 `\Chord`

Chords may be obtained by means of the `\Chord` command. It accepts an optional argument: the number of notes to step backwards —mostly for the `\Note` command— with 1 as default.

The best way to use it is to call the `\Note` command with the highest notes; then call the `\Chord` command; then call the `\Note` command with the duration 4 of a quarter note, to print them on top.

If the stem needs to be longer, the user may need the `\Enlarge` command, which accepts as arguments the 2 heights of both notes.

A musical staff in treble clef showing a chord of two notes. The first note is a whole note (duration 4) on G4. The second note is a quarter note (duration 4) on A4, with a longer stem. Below the staff, the commands used to create this notation are listed:

```

\Note{1}{6}      \Swap\Note{16}{7}
\Chord           \Chord
\Note{1}{4}      \Swap\Enlarge{7}{0}
                  \Note{4}{0}

```

For extreme uses of the command, extra `\Space` commands might be needed, as seen here:

`\Note{8}{5,4}` `\Note{8}{5,4}` `\Note{8}{5,4}`
`\Chord[2]` `\Chord` `\Chord[2]`
`\Note{4}{1,2}` `\Note{4}{2}` `\Note{4}{1}`
 `\Space[1]`

maybe fix
this

3.6 Keys and accidentals

3.6.1 `\Acc`

The `Acc` command prints an accidental *before* a note. It accepts 2 arguments.

The first argument is the type of accidental:

- -2 for the $\flat\flat$ double flat symbol.
- -1 for the $\flat$ flat symbol.
- 0 for the $\natural$ natural symbol.
- 1 for the $\sharp$ sharp symbol.
- 2 for the $\sharp\sharp$ double sharp symbol.

The second argument is the height of the symbol.

-2 -1 0 1 2

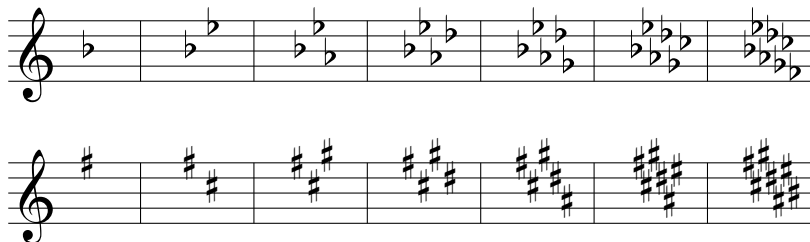
When there are several notes, the `\Accs` command might be useful. It accepts as an argument a list of pairs: type/height. Where there is no accidental, the spot shall be left empty.

`\Accs{-2/5, , -1/0}`
`\Note{5,3,0}`

3.6.2 \Key

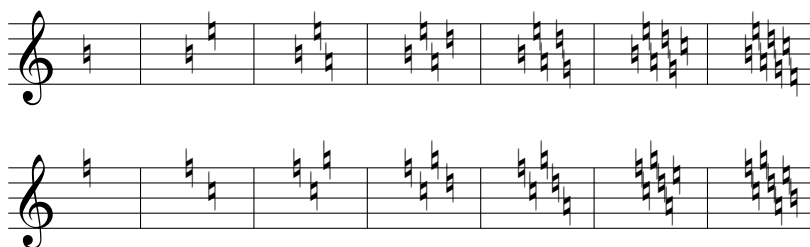
The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

It accepts an argument: a positive whole number from 1 to 7 for sharps $\sharp$, and a negative whole number from -1 to -7 for flats $\flat$.



3.6.3 \NoKey

The `\NoKey` command is the same as the `\Key` command, but it prints naturals $\natural$ instead of sharps or flats.



Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



```
\begin{mousik}
  \Clef{4}
  \Key{1}
  \TimeSig{C|}{ }

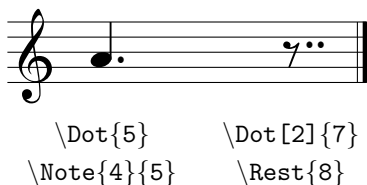
  \Acc{0}{8}
  \Note{1}{8}
  \Bar[|. ]
\end{mousik}
```


3.7 Other symbols: dots, ties, slurs and text

3.7.1 `\Dot`

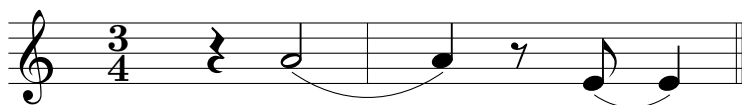
The `\Dot` command prints any amount of dots in front of a note or rest. It accepts an optional argument and a compulsory argument.

The default number of dots is 1, but it may be any whole number bigger than one. The second, compulsory argument is the height of the dot: `\Dot[2]{7}`.



3.7.2 `\Tied`

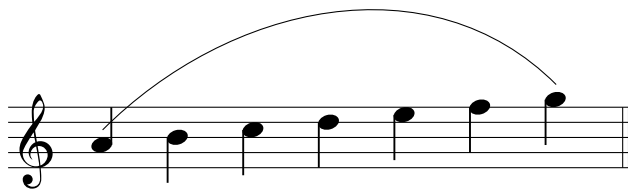
The `tied` environment draws a tie between the two notes placed inside. It has two arguments: the heights of both beginning and end notes (usually the same).



fix, add
swap

3.7.3 The slurred environment

The `slurred` environment draws a slur between the notes placed inside. It has two arguments: the heights of both beginning and end notes.



Fix angle

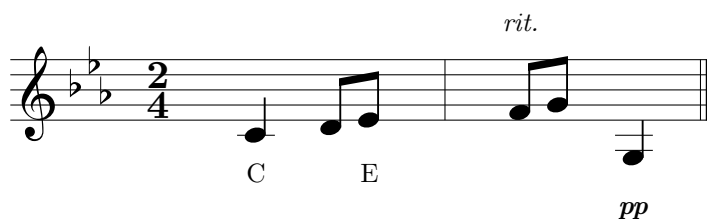
crescendo

articulation

3.8 `\Text` and `\Dynamics`

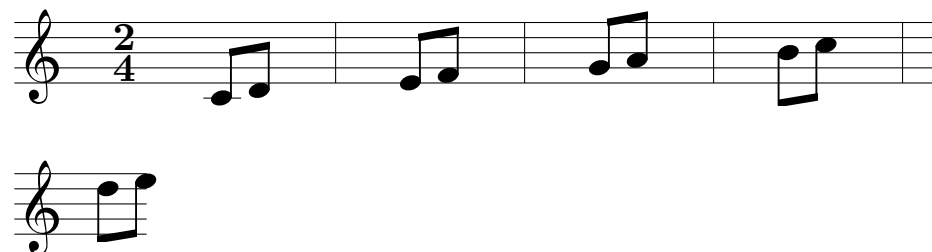
The `\Text` command allows the printing of any text string. It has a compulsory argument, but also an optional one where the user may choose the height of the string: `\Text[2]{\it rit.}`.

To write anything in the usual font of dynamics, the `\Dynamics` command will work the same: `\Dynamics[-0.5]{pp}`.



4 Observations

4.1 Overflow of lines



4.2 How do spaces work?

Spaces serve as the manual progression of an invisible cursor during the printing. A negative

explain
spaces

5 A very long and complete example

$\bullet = 62$

pp

5

6

3

f

Tempo command

6 The package code

beautiful
code

```

1  \RequirePackage[minimal]{leadsheets}
   \ useleadsheetslibraries {musicsymbols}
3  \RequirePackage{harmony}
   \RequirePackage{musixtex}
5
   \RequirePackage{tikz}
7
   %%%%%%%%%%%%%%
9  % Tools

11 \newcounter{Mevaluate}

13 \newcounter{Mmodulo}
   \newcommand{\MModulo}[2]{
15     \setcounter{Mmodulo}{#1}
       \setcounter{Mevaluate}{#2}
17     \addtocounter{Mevaluate}{-1}
       \loop
19     \ifnum\theMmodulo>\theMevaluate
       \addtocounter{Mmodulo}{-#2}
21     \repeat
       \ifnum\theMmodulo<0
23     \addtocounter{Mmodulo}{#2}
       \repeat
25 }

27 \newcounter{Mmax}
   \newcommand{\MMax}[2][\theMmax]{
29     \setcounter{Mmax}{#1}
       \ifnum#2>#1
31     \setcounter{Mmax}{#2}
       \fi
33 }

35 \newcounter{Mmin}
   \newcommand{\MMin}[2][\theMmin]{
37     \setcounter{Mmin}{#1}
       \ifnum#2<#1
39     \setcounter{Mmin}{#2}
       \fi
41 }

43 \newcounter{Mcount}
   \newcommand{\MCount}[1]{
45     \setcounter{Mcount}{0}
       \foreach\IND in {#1} {\stepcounter{Mcount}}
47 }

49 \newcommand{\MString}[3]{
   \ifthenelse {\equal{#1}{\string #2}}
51 {#3}{}
   }
53

```

```

55 % The main environment

57 \newcounter{Mfirst}
   \newcounter{MX}
59 \newcounter{MY}
   \newcounter{Mclef}
61 \newcounter{Mkey}
   \newcounter{Mindent}

63
65 \newcounter{Mlong}
   \setcounter{Mlong}{24}

67 \newenvironment{mousik}[1][0]
   {\setcounter{MX}{0}%
    \setcounter{MY}{0}%
69 \setcounter{Mclef}{0}%
    \setcounter{Mkey}{0}%
71 \setcounter{Mindent}{#1}%
    \noindent\begin{tikzpicture}\Space[#1]}
   {\addtocounter{MX}{-2}\MStaff
75 \setcounter{MX}{0}
    \setcounter{MY}{0}
77 \setcounter{Mclef}{0}
    \setcounter{Mkey}{0}
79 \setcounter{Mindent}{0}
    \end{tikzpicture}}

81
83 \newif\ifswap\swapfalse
   \newif\ifnoswap\noswaptrue

85 \newcommand{\Mswap}{
   \ifnoswap\noswapfalse
87 \else\noswaptrue
   \fi

89 }
   \newcommand{\Swap}{
91 \ifswap\swapfalse
   \else\swaptrue
93 \fi

95 }
   \newcommand{\MStaff}{
97 \foreach\INDi in {0,...,\theMY} {
   \foreach\INDj in {0,...,4} {
99 \ifnum\INDi=\theMY
   \ifnum\INDi=0
101 \draw (-0.25+\theMindent*0.5,-\INDi*2+\INDj*0.2)
   -- (\theMX*0.5,-\INDi*2+\INDj*0.2);
103 \else
   \draw (-0.25,-\INDi*2+\INDj*0.2)
105 -- (\theMX*0.5,-\INDi*2+\INDj*0.2);
   \fi
107 \else\ifnum\INDi=0
   \draw (-0.25+\theMindent*0.5,-\INDi*2+\INDj*0.2)
109 -- (\theMlong*0.5,-\INDi*2+\INDj*0.2);
   \else

```

```

111          \draw (-0.25,-\INDi*2+\INDj*0.2)
112          -- (\theMlong*0.5,-\INDi*2+\INDj*0.2);
113      \fi \fi
114  }
115 }
116 }
117
118 %\newcommand{\Clef}[1]{
119 %  \setcounter{Mclef}{#1}
120 %  \Adv
121 %
122 %  \ifnum#1=1
123 %    \node at (\theMX*0.5+0.1,-\theMY*2+0.15)
124 %    {{\resizebox{0.65cm}{0.8cm}{\trebleclef}}};
125 %  \fi \ifnum#1=2
126 %    \node at (\theMX*0.5+0.1,-\theMY*2+0.35)
127 %    {{\resizebox{0.65cm}{0.8cm}{\trebleclef}}};
128 %  \fi \ifnum#1=3
129 %    \node at (\theMX*0.5+0.1,-\theMY*2+0.42)
130 %    {{\resizebox{0.8cm}{0.6cm}{\bassclef}}};
131 %  \fi \ifnum#1=4
132 %    \node at (\theMX*0.5+0.1,-\theMY*2+0.62)
133 %    {{\resizebox{0.8cm}{0.6cm}{\bassclef}}};
134 %  \fi \ifnum#1=11
135 %    \node at (\theMX*0.5+0.1,-\theMY*2+0.42)
136 %    {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
137 %  \fi \ifnum#1=12
138 %    \node at (\theMX*0.5+0.1,-\theMY*2+0.62)
139 %    {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
140 %  \fi \ifnum#1=13
141 %    \node at (\theMX*0.5+0.1,-\theMY*2+0.82)
142 %    {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
143 %  \fi \ifnum#1=14
144 %    \node at (\theMX*0.5+0.1,-\theMY*2+1.02)
145 %    {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
146 %  \fi \ifnum#1=15
147 %    \node at (\theMX*0.5+0.1,-\theMY*2+1.22)
148 %    {{\resizebox{0.65cm}{0.55cm}{\altoclef}}};
149 %  \fi
150 %  \addtocounter{MX}{2}
151 %}
152
153 \newcommand{\Clef}[1]{
154   \setcounter{Mclef}{#1}
155   \Adv
156
157   \ifnum#1=1
158     \node at (\theMX*0.5+0.1,-\theMY*2+0.15)
159     {\trebleclef};
160   \fi \ifnum#1=2
161     \node at (\theMX*0.5+0.1,-\theMY*2+0.35)
162     {\trebleclef};
163   \fi \ifnum#1=3
164     \node at (\theMX*0.5+0.1,-\theMY*2+0.8)
165     {{\resizebox{0.7cm}{0.8cm}{\bassclef}}};
166   \fi \ifnum#1=4
167     \node at (\theMX*0.5+0.1,-\theMY*2+1)

```

```

    {{\resizebox{0.7cm}{0.8cm}{\bassclef}}};
169 \fi \ifnum#1=11
    \node at (\theMX*0.5+0.1,-\theMY*2+0.4)
171 {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
    \fi \ifnum#1=12
173 \node at (\theMX*0.5+0.1,-\theMY*2+0.6)
    {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
175 \fi \ifnum#1=13
    \node at (\theMX*0.5+0.1,-\theMY*2+0.8)
177 {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
    \fi \ifnum#1=14
179 \node at (\theMX*0.5+0.1,-\theMY*2+1)
    {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
181 \fi \ifnum#1=15
    \node at (\theMX*0.5+0.1,-\theMY*2+1.2)
183 {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
    \fi
185 \addtocounter{MX}{2}
}
187
\newlength{\MWidth}
189 \newlength{\MHeight}
\newcommand{\MResize}[1]{
191 \settowidth{\MWidth}{\#1}
    \settoheight{\MHeight}{\#1}
193 \resizebox{1.5\MWidth}{1.5\MHeight}{\#1}
}
195 \newcommand{\TimeSig}[2]{
    \Adv
197 \ifx\#2&
    \ifthenelse{\equal{\#1}{C}}{%
199 {\node at (\theMX*0.5,-\theMY*2+0.5) {\meterC};}%
    {\ifthenelse{\equal{\#1}{C}}{%
201 {\node at (\theMX*0.5,-\theMY*2+0.575) {\allabreve};}%
    {\node at (\theMX*0.5,-\theMY*2+0.4) {\bf \MResize{\#1}};}%
203 }
    \else
205 \node at (\theMX*0.5,-\theMY*2+0.6) {\bf \MResize{\#1}};
    \node at (\theMX*0.5,-\theMY*2+0.2) {\bf \MResize{\#2}};
207 \fi
    \addtocounter{MX}{3}
209 }
}
211 \newcommand{\Dot}[2][1]{
    \foreach \IND in {1,...,\#1}
213 {\node at (\theMX*0.5+0.1+0.15*\IND,-\theMY*2-0.21+0.1*\#2)
    {{\resizebox{0.1cm}{0.1cm}{\bullet$}}};}
215 }
}
217 \newcounter{Maux}
\newcounter{MauxB}
219 \newcommand{\Grupetto}[2][1]{
    \addtocounter{MX}{-1}
221 \setcounter{Mevaluate}{\theMX}
    \addtocounter{Mevaluate}{-\theMaux}
223 \setcounter{MauxB}{0}
    \ifnum\theMevaluate=2\else

```

```

225 \setcounter{MauxB}{21}
226 \fi
227 \node at (\theMaux*0.5+\theMevaluate*0.25-0.15-\theMauxB*0.01,-\theMY*2+#2*0.1)
228 {\ifx\theMevaluate\else#1\fi};
229 \addtocounter{MX}{1}
230 }
231 %%%%%%%%%%%
232 % Notes : bits
233
234 \newcommand{\Head}[1]{
235 \node [opacity=0.0] (tmp) at (\theMX*0.5,-\theMY*2-0.2+#1*0.1) {};
236 \fill [rotate around={16:(tmp.center)}] (\theMX*0.5-0.01,-\theMY*2-0.195+#1*0.1) ellipse (4pt and 2.75pt);
237
238 \newcommand{\HollowHead}[1]{
239 \node [opacity=0.0] (tmp) at (\theMX*0.5,-\theMY*2-0.2+#1*0.1) {};
240 \fill [black, rotate around={16:(tmp.center)}] (\theMX*0.5-0.01,-\theMY*2-0.195+#1*0.1) ellipse (4pt and 2.75pt);
241 \fill [white, rotate around={19:(tmp.center)}] (\theMX*0.5-0.01,-\theMY*2-0.195+#1*0.1) ellipse (3.2pt and 1.35pt);
242 }
243
244 \newcommand{\Menlarge}[3][0]{
245 \ifswap\Mswap\fi
246 \ifnoswap
247 \draw[line width=0.25mm] (\theMX*0.5+0.115,-\theMY*2+#2*0.1-0.2) --
248 (\theMX*0.5+0.115,-\theMY*2+#3*0.1+#1);
249 \else
250 \draw[line width=0.25mm] (\theMX*0.5-0.14,-\theMY*2+#2*0.1-0.2) --
251 (\theMX*0.5-0.14,-\theMY*2+#3*0.1-0.5-#1);
252 \fi
253 \ifswap\Mswap\fi
254 }
255 \newcommand{\Enlarge}[2]{
256 \Menlarge{#1}{#2}
257 \swapfalse
258 }
259
260 \newcommand{\Ledge}[1]{
261 \ifnum1>#1
262 \foreach\INDi in {0,...,#1} {
263 \ifthenelse {\isodd{\INDi}}{}%
264 {\draw (\theMX*0.5-0.25,-\theMY*2-0.2+\INDi*0.1) --
265 (\theMX*0.5+0.25,-\theMY*2-0.2+\INDi*0.1);}
266 }
267 \fi \ifnum1<#1
268 \foreach\INDi in {12,...,#1} {
269 \ifthenelse {\isodd{\INDi}}{}%
270 {\draw (\theMX*0.5-0.25,-\theMY*2-0.2+\INDi*0.1) --
271 (\theMX*0.5+0.25,-\theMY*2-0.2+\INDi*0.1);}
272 }
273 \fi
274 }
275
276 \newcommand{\Beam}[4]{
277 \ifswap\Mswap\fi
278 \ifnoswap
279 \draw[line width=1mm] (#1*0.5+0.103,-\theMY*2+#2*0.1#4) -- (\theMX*0.5-0.372,-\theMY*2+#3*0.1#4);
280 \else

```



```

\draw[line width=1mm] (#1*0.5-0.153,-\theMY*2+#2*0.1-0.5-#4) -- (\theMX*0.5-0.625,-\theMY*2+#3*0.1-0.
283 \fi
\ifswap\Mswap\fi
285 }

\newcommand{\BeamS}[1]{
\ifswap\Mswap\fi
287 \ifnoswap
\ifnum\theMfirst=\theMmin
289 \begin{scope}
\clip (\theMX*0.5-0.4,-\theMY*2+\theMmin*0.1-0.3) rectangle (\theMX*0.5+0.125,-\theMY*2+0.05+\theMmax*0.1-0.3);
291 \draw[line width=1mm] (\theMX*0.5-0.411,-\theMY*2+\theMmin*0.1-0.3) -- (\theMX*0.5+0.147,-\theMY*2+\theMmax*0.1-0.3);
293 \end{scope}
\else
295 \begin{scope}
\clip (\theMX*0.5-0.4,-\theMY*2+0.05+\theMmax*0.1-0.3) rectangle (\theMX*0.5+0.125,-\theMY*2+0.05+\theMmax*0.1-0.3);
297 \draw[line width=1mm] (\theMX*0.5-0.411,-\theMY*2+\theMmax*0.1-0.3) -- (\theMX*0.5+0.147,-\theMY*2+\theMmax*0.1-0.3);
299 \end{scope}
\fi
301 \else
\ifnum\theMfirst=\theMmin
303 \begin{scope}
\clip (\theMX*0.5-0.645,-\theMY*2+\theMmin*0.1-0.5-#1) rectangle (\theMX*0.5-0.129,-\theMY*2+\theMmax*0.1-0.5-#1);
305 \draw[line width=1mm] (\theMX*0.5-0.7,-\theMY*2+\theMmin*0.1-0.5-#1) -- (\theMX*0.5-0.1,-\theMY*2+\theMmax*0.1-0.5-#1);
\end{scope}
\else
307 \begin{scope}
\clip (\theMX*0.5-0.65,-\theMY*2+\theMmin*0.1-0.5-#1) rectangle (\theMX*0.5-0.129,-\theMY*2+\theMmax*0.1-0.5-#1);
309 \draw[line width=1mm] (\theMX*0.5-0.7,-\theMY*2+\theMmax*0.1-0.5-#1) -- (\theMX*0.5-0.1,-\theMY*2+\theMmax*0.1-0.5-#1);
311 \end{scope}
\fi
313 \fi
\ifswap\Mswap\fi
315 }

\newcommand{\Flag}[2][0]{
\ifswap\Mswap\fi
317 \ifnoswap
\begin{scope}
319 \clip (\theMX*0.5+0.13,-\theMY*2+#2*0.1-0.75+#1) rectangle (\theMX*0.5+0.35,-\theMY*2+#2*0.1+0.05+#1);
\draw (\theMX*0.5+0.05,-\theMY*2-0.3+#2*0.1+#1) {\resizebox{0.55cm}{0.65cm}{\Acht}};
321 \end{scope}
\fill [white] (\theMX*0.5+0.205,-\theMY*2-0.308+#2*0.1+#1) circle (1.75pt);
323 \else
\begin{scope}
325 \node [opacity=0.0] (tmp) at (\theMX*0.5,-\theMY*2+#2*0.1) {};
\clip (\theMX*0.5-0.125,-\theMY*2+#2*0.1-0.55-#1) rectangle (\theMX*0.5+0.09,-\theMY*2+#2*0.1+0.15-#1);
327 \node[rotate around={180:(tmp.center)}] at (\theMX*0.5+0.175,-\theMY*2+0.165+#2*0.1+#1) {\reflectbox{\resizebox{0.55cm}{0.65cm}{\Acht}}};
329 \end{scope}
\fill [white] (\theMX*0.5-0.035,-\theMY*2-0.161+#2*0.1-#1) circle (2pt);
331 \fi
\ifswap\Mswap\fi
333 }

\newcommand{\Semibreve}[1]{
335 \node at (\theMX*0.5,-\theMY*2-0.2+#1*0.1) {\resizebox{0.3cm}{0.2cm}{\Ganz}};
337 }

```

```

339 \newcommand{\Minim}[1]{
341   \HollowHead{#1}
341   \ifnum#1<6\else
343     \Mswap
343   \fi
345   \Menlarge[0.3]{#1}{#1}
345 }
347 \newcommand{\Crotchet}[1]{
347   \Head{#1}
349   \ifnum#1<6\else
349     \Mswap
351   \fi
351   \Menlarge[0.3]{#1}{#1}
353 }
353 \newcommand{\Quaver}[1]{
355   \Head{#1}
357   \ifnum#1<6\else
357     \Mswap
359   \fi
359   \Menlarge[0.4]{#1}{#1}
361   \Flag[0.4]{#1}
361 }
363 \newcommand{\Semiquaver}[1]{
363   \Head{#1}
365   \ifnum#1<6\else
365     \Mswap
367   \fi
367   \Menlarge[0.5]{#1}{#1}
369   \Flag[0.5]{#1}
369   \Flag[0.35]{#1}
371 }
371 \newcommand{\Demisemiquaver}[1]{
373   \Head{#1}
373   \ifnum#1<6\else
375     \Mswap
375   \fi
377   \Menlarge[0.6]{#1}{#1}
377   \Flag[0.6]{#1}
379   \Flag[0.45]{#1}
379   \Flag[0.3]{#1}
381 }
381 %%%%%%%%%%%%%%%
383 % Notes : everything together

385 \newcounter{Mduration}
385 \newcommand{\Note}[2]{
387   \Adv\noswaptrue
387   \setcounter{Maux}{\theMX}
389
389   \ifnum#1=1
391     \foreach\IND in {#2} {\Semibreve{\IND}\stepcounter{MX}}
391   \else\ifnum#1=2
393     \foreach\IND in {#2} {\Minim{\IND}\stepcounter{MX}}
393   \else\ifnum#1=4
395     \foreach\IND in {#2} {\Crotchet{\IND}\stepcounter{MX}}

```

```

\ else
397 \MCount{#2}
\ifnum\theMcount=1
399 \ifnum#1=8 \Quaver{#2} \else
\ifnum#1=16 \Semiquaver{#2} \else
401 \ifnum#1=32 \Demisemiquaver{#2} \fi\fi\fi
\else\ifnum\theMcount=2
403 \MTwoNote{#1}{#2}
\else\ifnum\theMcount>2
405 \MFlatNote{#1}{#2}
\fi\fi\fi
407 \fi\fi\fi

409 \setcounter{MX}{\theMaux}
\foreach\IND in {#2} {
411 \Ledger{\IND}\stepcounter{MX}}
\addtocounter{MX}{1}
413 \swapfalse
}

415 \newcounter{MMdur}
417 \newcommand{\MMixedNote}[2]{
% \foreach\INDd in {#1} {
419 % \Max{\INDd}
% }
421 % \setcounter{MMdur}{\theMMax}
% \MHeads{#2}
423 %
% \setcounter{Mfirst}{-1}
425 % \ifnum\theMMdur=16
% \foreach \INDi
427 % {}{}
% \fi
429 }

431 \newcommand{\MHeads}[1]{
\setcounter{Mfirst}{-50}
433 \setcounter{Mmax}{-50}
\setcounter{Mmin}{50}
435 \foreach\IND in {#1} {
\Head{\IND}\stepcounter{MX}
437 \MMax{\IND}\MMin{\IND} % max, min, first
\ifnum\theMfirst=-50\setcounter{Mfirst}{\IND}\fi
439 }
}

441
441 \newcommand{\MTwoNote}[2]{
443 \MHeads{#2}
\ifnum\theMmax<6\else\Mswap\fi
445 \setcounter{MX}{\theMaux}
\foreach\IND in {#2} {
447 \ifnum#1<16
\Menlarge[0.4]{\IND}{\IND}
449 \fi\ifnum#1=16
\Menlarge[0.5]{\IND}{\IND}
451 \fi\ifnum#1=32
\Menlarge[0.6]{\IND}{\IND}

```

```

453     \ fi
454     \stepcounter{MX}
455   }
456   \addtocounter{MX}{-1}
457   \ifnum#1=8
458     \BeamS{+0.4}
459   \ fi \ifnum#1=16
460     \BeamS{+0.5}
461     \BeamS{+0.35}
462   \ fi \ifnum#1=32
463     \BeamS{+0.6}
464     \BeamS{+0.45}
465     \BeamS{+0.3}
466   \ fi
467 }
468 \newcommand{\MFlatNote}[2]{
469   \MHeads{#2}
470   \ifnum\theMmax<6
471     \ifswap\setcounter{Mevaluate}{\theMmin}
472     \else\setcounter{Mevaluate}{\theMmax}
473   \ fi
474   \else\Mswap
475     \ifswap\setcounter{Mevaluate}{\theMmax}
476     \else\setcounter{Mevaluate}{\theMmin}
477   \ fi
478   \ fi
479   \setcounter{MX}{\theMaux}
480   \foreach\IND in {#2} {
481     \ifnum#1<16
482       \Menlarge[0.4]{\IND}{\theMevaluate}
483     \ fi \ifnum#1=16
484       \Menlarge[0.5]{\IND}{\theMevaluate}
485     \ fi \ifnum#1=32
486       \Menlarge[0.6]{\IND}{\theMevaluate}
487     \ fi
488     \stepcounter{MX}
489   }
490   \ifnum#1=8
491     \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.4}
492   \ fi \ifnum#1=16
493     \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.5}
494     \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.35}
495   \ fi \ifnum#1=32
496     \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.6}
497     \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.45}
498     \Beam{\theMaux}{\theMevaluate}{\theMevaluate}{+0.3}
499   \ fi
500 }
501 }
502
503 \newcommand{\Rest}[1]{
504   \Adv
505   \ifnum#1=1
506     \node at (\theMX*0.5,-\theMY*2+0.52)
507       {\resizebox{0.3cm}{0.15cm}{\GaPa}};
508   \ fi \ifnum#1=2

```

```

\node at (\theMX*0.5,-\theMY*2+0.42)
511 {\resizebox{0.3cm}{0.15cm}{\HaPa}};
\fi \ifnum#1=4
513 \node at (\theMX*0.5,-\theMY*2+0.4)
{\resizebox{0.3cm}{0.7cm}{\ViPa}};
\fi \ifnum#1=8
515 \node at (\theMX*0.5,-\theMY*2+0.32)
517 {\resizebox{0.3cm}{0.45cm}{\AcPa}};
\fi \ifnum#1=16
519 \node at (\theMX*0.5,-\theMY*2+0.3)
{\resizebox{0.35cm}{0.525cm}{\SePa}};
521 \fi \ifnum#1=32
\node at (\theMX*0.5,-\theMY*2+0.375)
523 {\resizebox{0.4cm}{0.6cm}{\ZwPa}};
\fi
525
\addtocounter{MX}{2}
527 }

529 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

531 \newcommand{\MAcc}[3][-0.25]{
\ifnum#2=0
533 \node at (\theMX*0.5#1,-\theMY*2-0.2+0.1*#3)
{{\resizebox{0.15cm}{0.35cm}{\natural}}};
535 \fi \ifnum#2=1
\node at (\theMX*0.5#1,-\theMY*2-0.2+0.1*#3)
537 {{\resizebox{0.15cm}{0.35cm}{\sharp}}};
\fi \ifnum#2=-1
539 \node at (\theMX*0.5#1,-\theMY*2-0.1+0.1*#3)
{{\resizebox{0.15cm}{0.35cm}{\flat}}};
541 \fi \ifnum#2=2
\node at (\theMX*0.5#1,-\theMY*2-0.205+0.1*#3)
543 {{\resizebox{0.15cm}{0.15cm}{\doublesharp}}};
\fi \ifnum#2=-2
545 \node at (\theMX*0.5#1,-\theMY*2-0.1+0.1*#3)
{{\resizebox{0.2cm}{0.35cm}{\doubleflat}}};
547 \fi
}
549 \newcommand{\Acc}[2]{\MAcc{#1}{#2}}
\newcommand{\Accs}[1]{
551 \setcounter{Maux}{\theMX}
\foreach \INDi/\INDj in {#1} {
553 \ifthenelse {\equal{\INDi}{}}{}%
{\MAcc{\INDi}{\INDj}}
555 \stepcounter{MX}
}
557 \setcounter{MX}{\theMaux}
}

559
\newcommand{\MAccFN}[2][-0.25]{
561 \KB{#2}
\MAcc{#1}{0}{\theMaux}}
563 \newcommand{\MAccSN}[2][-0.25]{
\KS{#2}
565 \MAcc{#1}{0}{\theMaux}}
\newcommand{\MAccF}[2][-0.25]{

```

```

567     \KB{#2}
        \MAcc[#1]{-1}{\theMaux}}
569 \newcommand{\MAccS}[2][-0.25]{
        \KS{#2}
571     \MAcc[#1]{1}{\theMaux}}

573 \newcommand{\KB}[1]{
        \setcounter{Maux}{#1}
575     \addtocounter{Maux}{-3}

577     \ifnum\theMclef=1
        \addtocounter{Maux}{-2}
579     \fi \ifnum\theMclef=2
        \addtocounter{Maux}{0}
581     \fi \ifnum\theMclef=3
        \addtocounter{Maux}{3}
583     \fi \ifnum\theMclef=4
        \addtocounter{Maux}{-2}
585     \fi \ifnum\theMclef=11
        \addtocounter{Maux}{2}
587     \fi \ifnum\theMclef=12
        \addtocounter{Maux}{-3}
589     \fi \ifnum\theMclef=13
        \addtocounter{Maux}{-1}
591     \fi \ifnum\theMclef=14
        \addtocounter{Maux}{1}
593     \fi \ifnum\theMclef=15
        \addtocounter{Maux}{3}
595     \fi
        \MModulo{\theMaux}{7}
597     \setcounter{Maux}{\theMmodulo}
        \addtocounter{Maux}{3}
599 }
\newcommand{\KS}[1]{
601     \setcounter{Maux}{#1}
        \addtocounter{Maux}{-5}
603
        \ifnum\theMclef=1
        \addtocounter{Maux}{-2}
605     \fi \ifnum\theMclef=2
        \addtocounter{Maux}{0}
607     \fi \ifnum\theMclef=3
        \addtocounter{Maux}{3}
609     \fi \ifnum\theMclef=4
        \addtocounter{Maux}{-2}
611     \fi \ifnum\theMclef=11
        \addtocounter{Maux}{2}
613     \fi \ifnum\theMclef=12
        \addtocounter{Maux}{-3}
615     \fi \ifnum\theMclef=13
        \addtocounter{Maux}{-1}
617     \fi \ifnum\theMclef=14
        \addtocounter{Maux}{1}
619     \fi \ifnum\theMclef=15
        \addtocounter{Maux}{3}
621     \fi
623     \MModulo{\theMaux}{7}

```

```

        \setcounter{Maux}{\theMmodulo}
625     \addtocounter{Maux}{5}
    }
627 \newcommand{\Key}[1]{
629     \setcounter{Mkey}{#1}
        \ifnum#1=-1
631         \MAccF[-0.2]{6}
        \fi \ifnum#1=-2
633         \MAccF[-0.35]{6}
        \MAccF[-0.05]{9}
635         \fi \ifnum#1=-3
        \MAccF[-0.4]{6}
637         \MAccF[-0.15]{9}
        \MAccF[-0.0]{5}
639         \fi \ifnum#1=-4
        \MAccF[-0.45]{6}
641         \MAccF[-0.25]{9}
        \MAccF[-0.1]{5}
643         \MAccF[+0.1]{8}
        \fi \ifnum#1=-5
645         \MAccF[-0.45]{6}
        \MAccF[-0.25]{9}
647         \MAccF[-0.15]{5}
        \MAccF[+0.05]{8}
649         \MAccF[+0.15]{4}
        \fi \ifnum#1=-6
651         \MAccF[-0.45]{6}
        \MAccF[-0.3]{9}
653         \MAccF[-0.2]{5}
        \MAccF[-0.05]{8}
655         \MAccF[+0.05]{4}
        \MAccF[+0.2]{7}
657         \fi \ifnum#1=-7
        \MAccF[-0.45]{6}
659         \MAccF[-0.35]{9}
        \MAccF[-0.25]{5}
661         \MAccF[-0.15]{8}
        \MAccF[-0.05]{4}
663         \MAccF[+0.05]{7}
        \MAccF[+0.15]{3}
665         \fi
        \ifnum#1=1
667         \MAccS[-0.2]{10}
        \fi \ifnum#1=2
669         \MAccS[-0.35]{10}
        \MAccS[-0.05]{7}
671         \fi \ifnum#1=3
        \MAccS[-0.425]{10}
673         \MAccS[-0.2]{7}
        \MAccS[-0.025]{11}
675         \fi \ifnum#1=4
        \MAccS[-0.45]{10}
677         \MAccS[-0.25]{7}
        \MAccS[-0.1]{11}
679         \MAccS[+0.1]{8}
        \fi \ifnum#1=5

```

```

681 \MAccS[-0.45]{10}
    \MAccS[-0.3]{7}
683 \MAccS[-0.15]{11}
    \MAccS[-0.0]{8}
685 \MAccS[+0.15]{5}
    \fi \ifnum#1=6
687 \MAccS[-0.4]{10}
    \MAccS[-0.3]{7}
689 \MAccS[-0.2]{11}
    \MAccS[-0.1]{8}
691 \MAccS[+0.0]{5}
    \MAccS[+0.1]{9}
693 \fi \ifnum#1=7
    \MAccS[-0.45]{10}
695 \MAccS[-0.35]{7}
    \MAccS[-0.25]{11}
697 \MAccS[-0.15]{8}
    \MAccS[-0.05]{5}
699 \MAccS[+0.05]{9}
    \MAccS[+0.15]{6}
701 \fi
    \stepcounter{MX}
703 }

705 \newcommand{\NoKey}[1]{
    \setcounter{Mkey}{0}
707 \ifnum#1=-1
    \MAccFN[-0.2]{6}
709 \fi \ifnum#1=-2
    \MAccFN[-0.35]{6}
711 \MAccFN[-0.05]{9}
    \fi \ifnum#1=-3
713 \MAccFN[-0.4]{6}
    \MAccFN[-0.15]{9}
715 \MAccFN[-0.0]{5}
    \fi \ifnum#1=-4
717 \MAccFN[-0.45]{6}
    \MAccFN[-0.25]{9}
719 \MAccFN[-0.1]{5}
    \MAccFN[+0.1]{8}
721 \fi \ifnum#1=-5
    \MAccFN[-0.45]{6}
723 \MAccFN[-0.25]{9}
    \MAccFN[-0.15]{5}
725 \MAccFN[+0.05]{8}
    \MAccFN[+0.15]{4}
727 \fi \ifnum#1=-6
    \MAccFN[-0.45]{6}
729 \MAccFN[-0.3]{9}
    \MAccFN[-0.2]{5}
731 \MAccFN[-0.05]{8}
    \MAccFN[+0.05]{4}
733 \MAccFN[+0.2]{7}
    \fi \ifnum#1=-7
735 \MAccFN[-0.45]{6}
    \MAccFN[-0.35]{9}
737 \MAccFN[-0.25]{5}

```



```

739     \MAccFN[-0.15]{8}
       \MAccFN[-0.05]{4}
       \MAccFN[+0.05]{7}
741     \MAccFN[+0.15]{3}
       \fi
743     \ifnum#1=1
       \MAccSN[-0.2]{10}
745     \fi \ifnum#1=2
       \MAccSN[-0.35]{10}
747     \MAccSN[-0.05]{7}
       \fi \ifnum#1=3
749     \MAccSN[-0.425]{10}
       \MAccSN[-0.2]{7}
751     \MAccSN[-0.025]{11}
       \fi \ifnum#1=4
753     \MAccSN[-0.45]{10}
       \MAccSN[-0.25]{7}
755     \MAccSN[-0.1]{11}
       \MAccSN[+0.1]{8}
757     \fi \ifnum#1=5
       \MAccSN[-0.45]{10}
759     \MAccSN[-0.3]{7}
       \MAccSN[-0.15]{11}
761     \MAccSN[-0.0]{8}
       \MAccSN[+0.15]{5}
763     \fi \ifnum#1=6
       \MAccSN[-0.4]{10}
765     \MAccSN[-0.3]{7}
       \MAccSN[-0.2]{11}
767     \MAccSN[-0.1]{8}
       \MAccSN[+0.0]{5}
769     \MAccSN[+0.1]{9}
       \fi \ifnum#1=7
771     \MAccSN[-0.45]{10}
       \MAccSN[-0.35]{7}
773     \MAccSN[-0.25]{11}
       \MAccSN[-0.15]{8}
775     \MAccSN[-0.05]{5}
       \MAccSN[+0.05]{9}
777     \MAccSN[+0.15]{6}
       \fi
779     \stepcounter{MX}
}

781     %%%%%%%%%%%
783
       \newcommand{\Text}[2][0]{
785     \node at (\theMX*0.5,-\theMY*2-0.7+#1) {\#2};
       }

787
       \newcommand{\muskern}{\kern-.15ex }
789
       \makeatletter
       \newcommand\dynmark[1]{\normalfont\bfseries\itshape
791         \@tfor\next:=#1\do{\put@muskern\next}\}
       \newcommand{\put@muskern}{\let\put@muskern\muskern}
793     \makeatother
       \newcommand{\Dynamics}[2][0]{\Text[#1]{\dynmark{\#2}}}

```

```

795 \newcommand{\Chord}[1][1]{
797   \addtocounter{MX}{-1}
   \addtocounter{MX}{-#1}
799 \newcommand{\Space}[1][2]{\Adv\addtocounter{MX}{#1}}

801 \newcommand{\Bar}[1][1]{
   \ifnum\theMX>\theMlong
803   \addtocounter{MX}{-1}
   \fi
805   \MString{#1}{|}
   {\draw (\theMX*0.5,-\theMY*2+0.8) -- (\theMX*0.5,-\theMY*2);}
807   \MString{#1}{||}
   {\draw (\theMX*0.5-0.06,-\theMY*2+0.8) -- (\theMX*0.5-0.06,-\theMY*2);
809   \draw (\theMX*0.5+0.06,-\theMY*2+0.8) -- (\theMX*0.5+0.06,-\theMY*2);}
   \MString{#1}{|.}
811   {\draw (\theMX*0.5-0.12,-\theMY*2+0.8) -- (\theMX*0.5-0.12,-\theMY*2);
   \draw[line width=1mm] (\theMX*0.5,-\theMY*2+0.805) -- (\theMX*0.5,-\theMY*2-0.005);}
813   \MString{#1}{|:}
   {\node at (\theMX*0.5-0.24,0.4-\theMY*2) {\resizebox{0.175cm}{0.25cm}{:}};
815   \draw (\theMX*0.5-0.12,-\theMY*2+0.8) -- (\theMX*0.5-0.12,-\theMY*2);
   \draw[line width=1mm] (\theMX*0.5,-\theMY*2+0.805) -- (\theMX*0.5,-\theMY*2-0.005);}
817   \MString{#1}{|::}
   {\draw[line width=1mm] (\theMX*0.5,-\theMY*2+0.805) -- (\theMX*0.5,-\theMY*2-0.005);
819   \node at (\theMX*0.5+0.24,0.4-\theMY*2) {\resizebox{0.175cm}{0.25cm}{:}};
   \draw (\theMX*0.5+0.12,-\theMY*2+0.8) -- (\theMX*0.5+0.12,-\theMY*2);}
821   \MString{#1}{|:::}
   {\node at (\theMX*0.5-0.24,0.4-\theMY*2) {\resizebox{0.175cm}{0.25cm}{:}};
823   \draw (\theMX*0.5-0.12,-\theMY*2+0.8) -- (\theMX*0.5-0.12,-\theMY*2);
   \draw[line width=1mm] (\theMX*0.5,-\theMY*2+0.805) -- (\theMX*0.5,-\theMY*2-0.005);
825   \node at (\theMX*0.5+0.24,0.4-\theMY*2) {\resizebox{0.175cm}{0.25cm}{:}};
   \draw (\theMX*0.5+0.12,-\theMY*2+0.8) -- (\theMX*0.5+0.12,-\theMY*2);}
827   \addtocounter{MX}{2}
   }

829 \newcommand{\Adv}{
831   \addtocounter{Mlong}{-1}
   \ifnum\theMX>\theMlong
833     \addtocounter{MY}{1}
     \setcounter{MX}{0}
835     \ifnum\theMclef=0\else
     \Clef{\theMclef}
837     \fi
     \ifnum\theMkey=0\else
839     \Key{\theMkey}
     \fi
841   \fi
   \addtocounter{Mlong}{1}
843 }

845 \newcounter{Mtied}
\newcounter{MtiedA}
847 \newcounter{MtiedB}
\newenvironment{tied}[2]
849 {\setcounter{Mtied}{\theMX}\setcounter{MtiedA}{#1}\setcounter{MtiedB}{#2}}
{\draw (\theMtied*0.5,-\theMY*2+0.1*\theMtiedA-0.35) to[out=-35,in=-145] (-1+\theMX*0.5,-\theMY*2+0.1*\theMtiedB-
851 %{\draw (\theMtied*0.5,-\theMY*2+0.1*\theMtiedA) to[out=45,in=135] (-1+\theMX*0.5,-\theMY*2+0.1*\theMtiedB);}

```

```

853 \newcommand{\Tie}[2][1]{
      \draw (-1+\theMX*0.5,-\theMY*2+0.1*#2-0.35) to[out=-35,in=-145] (-0.5+#1*0.5+\theMX*0.5,-\theMY*2+0.1*#2-0.35);
855 }

857 \newcounter{Mslurred}
      \newcounter{MslurredA}
859 \newcounter{MslurredB}
      \newenvironment{slurred}[2]
861 {\setcounter{Mslurred}{\theMX}\setcounter{MslurredA}{#1}\setcounter{MslurredB}{#2}}
      {\draw (\theMslurred*0.5,-\theMY*2+0.1*\theMslurredA) to[out=45,in=135] (-1+\theMX*0.5,-\theMY*2+0.1*\theMslurredB);
863 %{\draw (\theMslurred*0.5,-\theMY*2+0.1*\theMslurredA-0.35) to[out=-45,in=-135] (-1+\theMX*0.5,-\theMY*2+0.1*\theMslurredB);

```